

Spatial synchrony stacks: Phase 1.5

What the validated stack means, which diagnostics survive, and the bounded Phase 2 decision

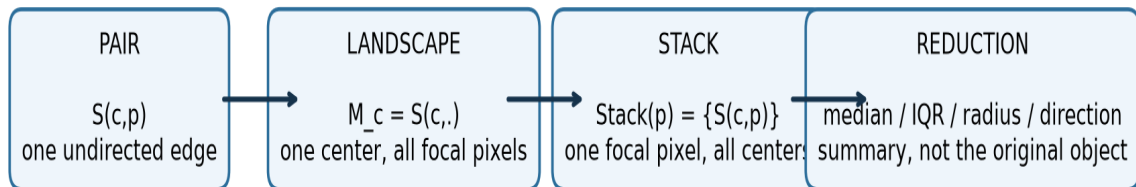
Decision: GO WITH CHANGES for the 100 by 100 bounded benchmark. Retain separate cold, warm, and Delta pair values; robust focal magnitude/heterogeneity; nested radius rings; and a multidimensional adjacent-panel signature. Do not promote a universal radius, KDE regime map, or a scalar Q.

Evidence base

Input	Exact Phase 1 observed experiment
Artifact	artifacts/synchrony-stack-phase1/prism_20x20_synchrony_stack.nc
Fingerprint	sha256:4b4a3c30fbd5ccb2a4f8b5cefc2ca2b127e7bf31d646b27091b354b447df9acb
Repository base commit	a852137944cabf238e9321f2eecf09e483c9b9f5
Working-tree qualifier	base commit plus uncommitted Phase 1/1.5 changes
PRISM interval	2023-11-01T00:00:00.000000000 through 2024-01-30T00:00:00.000000000 (91 labels)
Grid / centers / pairs	20 by 20 / 400 / 80,200 canonical pairs
Settings	90-day window; min_t=10; median split q=0.5
Climate recomputation	None in Phase 1.5

1. Scientific hierarchy

Phase 1 hierarchy: related views of the same pairwise field, not interchangeable quantities

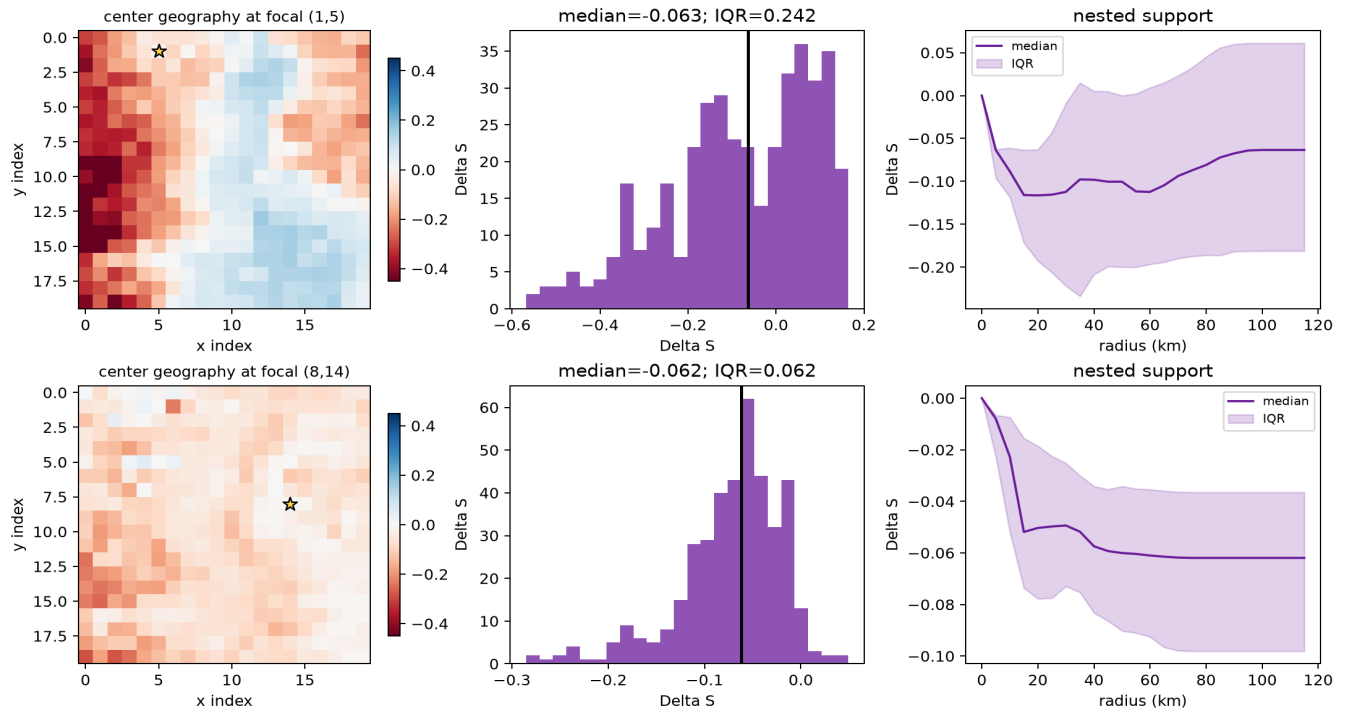


Panel comparison $Q(A,B)$ is second-order: it compares two complete edge patterns.

A focal stack is the set of incident edge values at one node. A center landscape is the same undirected edge field viewed outward from a center. Panel comparison is second-order: it compares two vectors of incident edge patterns. Pair calculation benefits from graph symmetry and sparse finite-radius storage; focal summaries, radius rings, direction bins, and maps remain clearer as raster operations.

2. Median alone fails

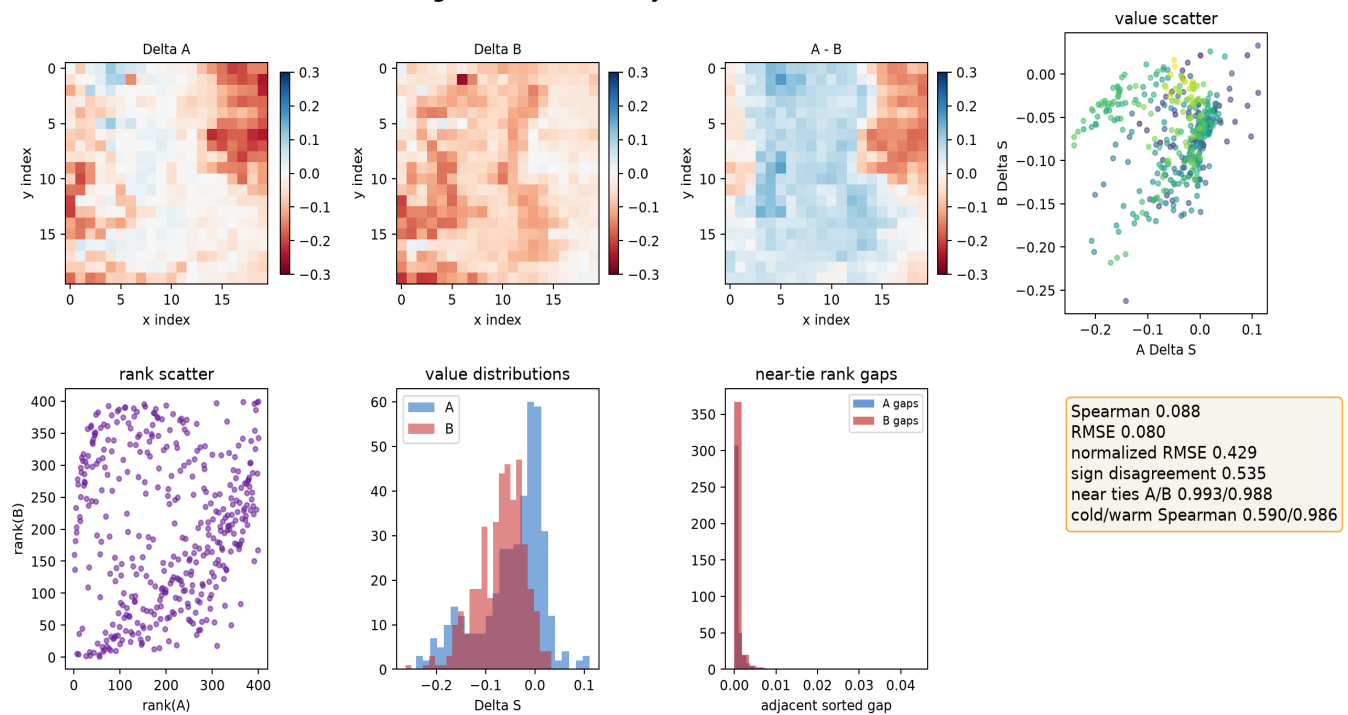
Similar medians can hide radically different stack structure



The two real focal pixels differ in median by only 0.0015, yet their IQRs are approximately 0.242 and 0.062. A median map therefore needs at least one robust heterogeneity field.

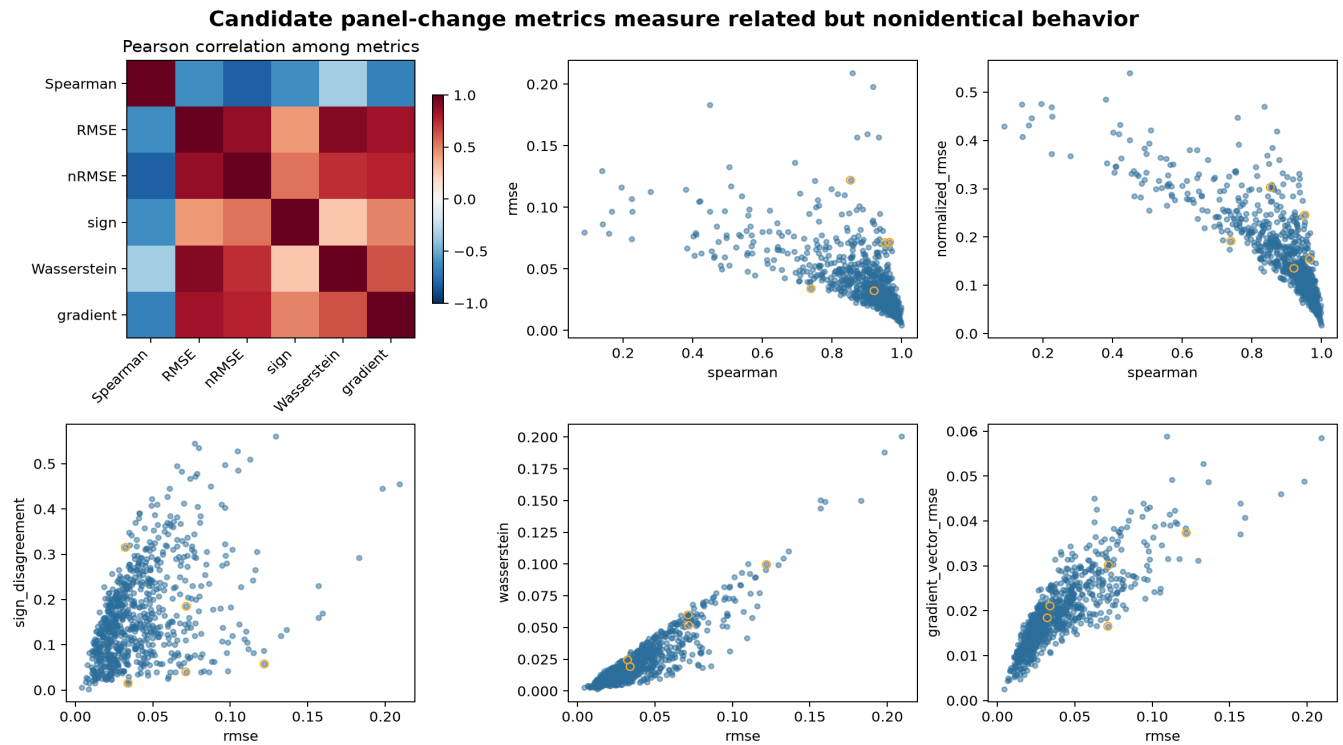
3. Why the strongest Spearman break looks modest

Strongest rank break: adjacent centers 132 and 133



$Q=0.088$ is not a software error. The two Delta panels have narrow robust ranges (about 0.209 and 0.151) and almost every value has a neighbor within 0.005, so modest perturbations reorder many pixels. Absolute RMSE is 0.080, normalized RMSE is 0.429, and deadbanded sign disagreement is 0.535. Warm ranks remain very stable (0.986); cold changes dominate (cold Spearman 0.590). The evidence supports rank instability plus real local/sign change, not a wholesale spatial translation or a single causal boundary.

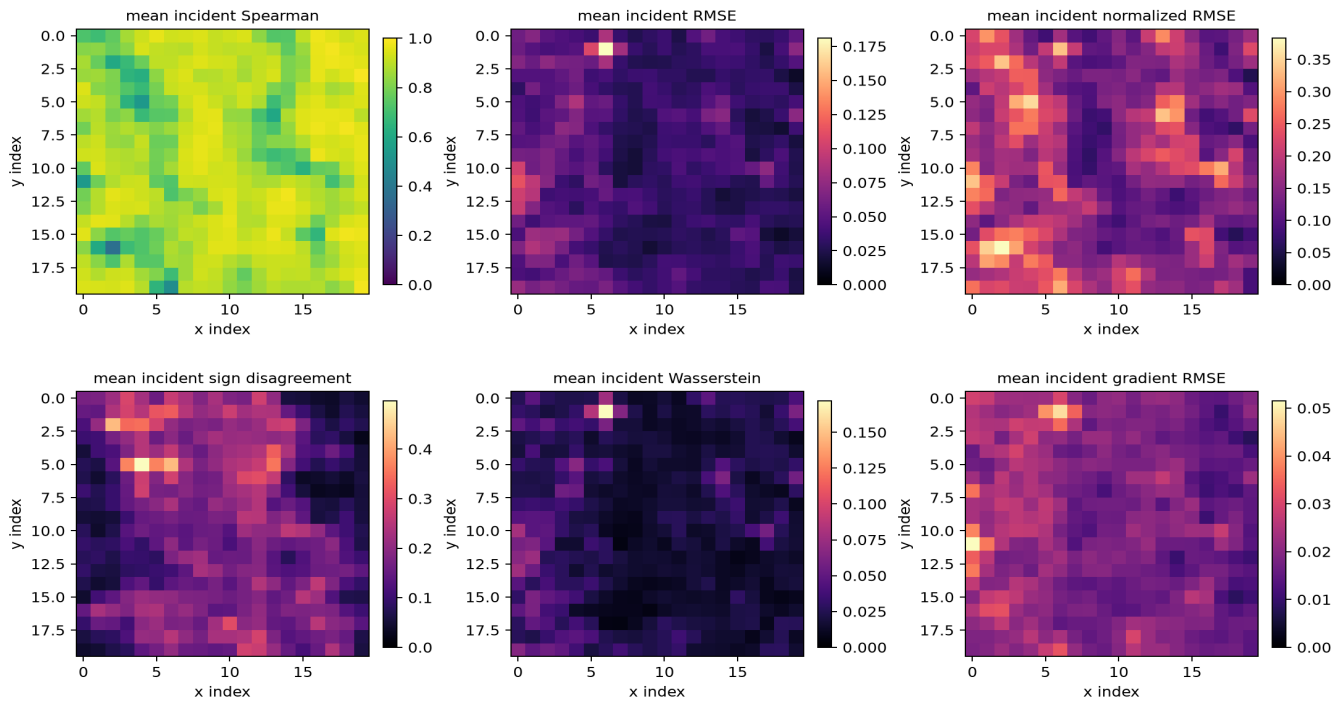
4. Panel-change metrics



Spearman is useful for rank structure but unsafe alone. Retain a compact signature: normalized or absolute magnitude change, rank structure with dynamic-range context, deadbanded sign change, and gradient-vector change. Wasserstein ignores geography and is strongly redundant with RMSE in this block.

5. Where panel change occurs

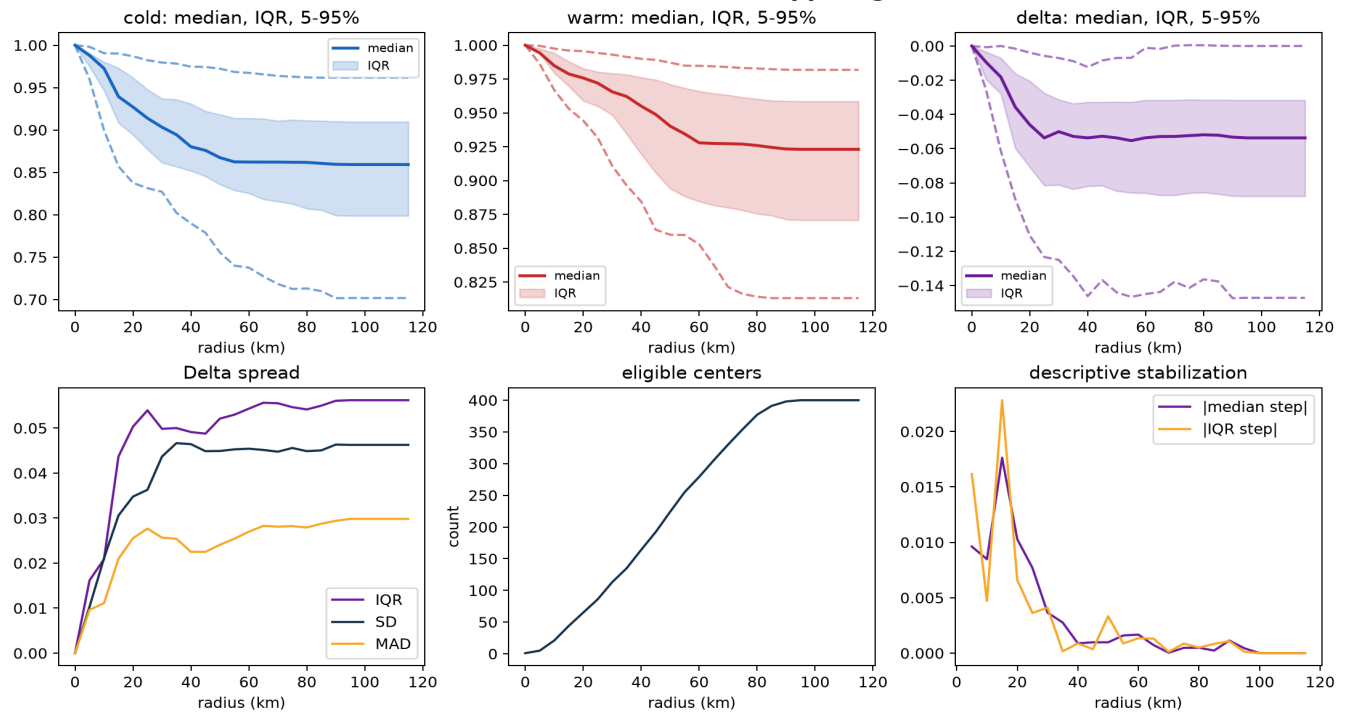
Panel change is attached to center-landscape transitions



Each map averages comparisons incident to a center. These are properties of landscape transitions, not focal-stack reductions.

6. Low-IQR radius growth

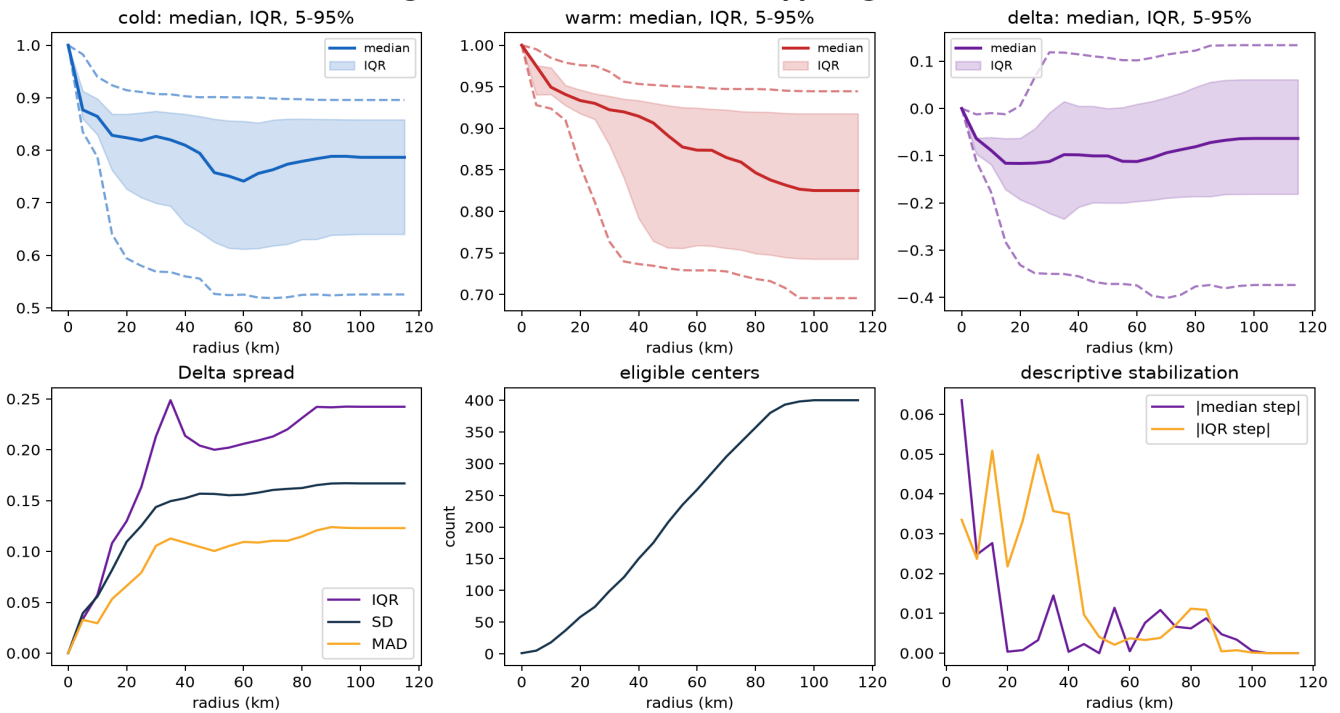
low-IQR focal (17,5): nested support growth



At focal (17,5), Delta IQR is about 0.056 and the experimental median/IQR stable radii are both 15 km. This is a narrow, comparatively uniform stack in this bounded window.

7. High-IQR radius growth

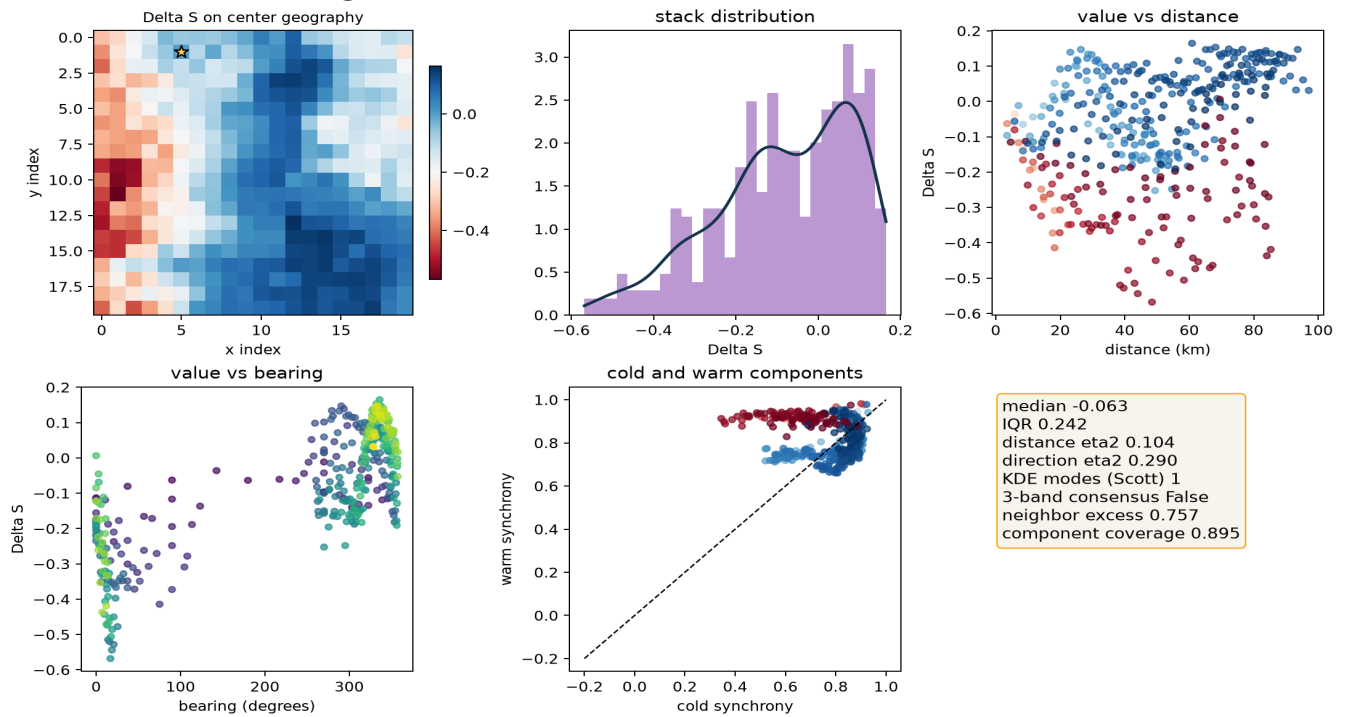
high-IQR focal (1,5): nested support growth



At focal (1,5), Delta IQR is about 0.242. The experimental median and IQR criteria do not settle until about 80 km. Curves can be nonmonotonic; the criterion is descriptive, not a validated synchrony radius.

8. The high-IQR focal stack

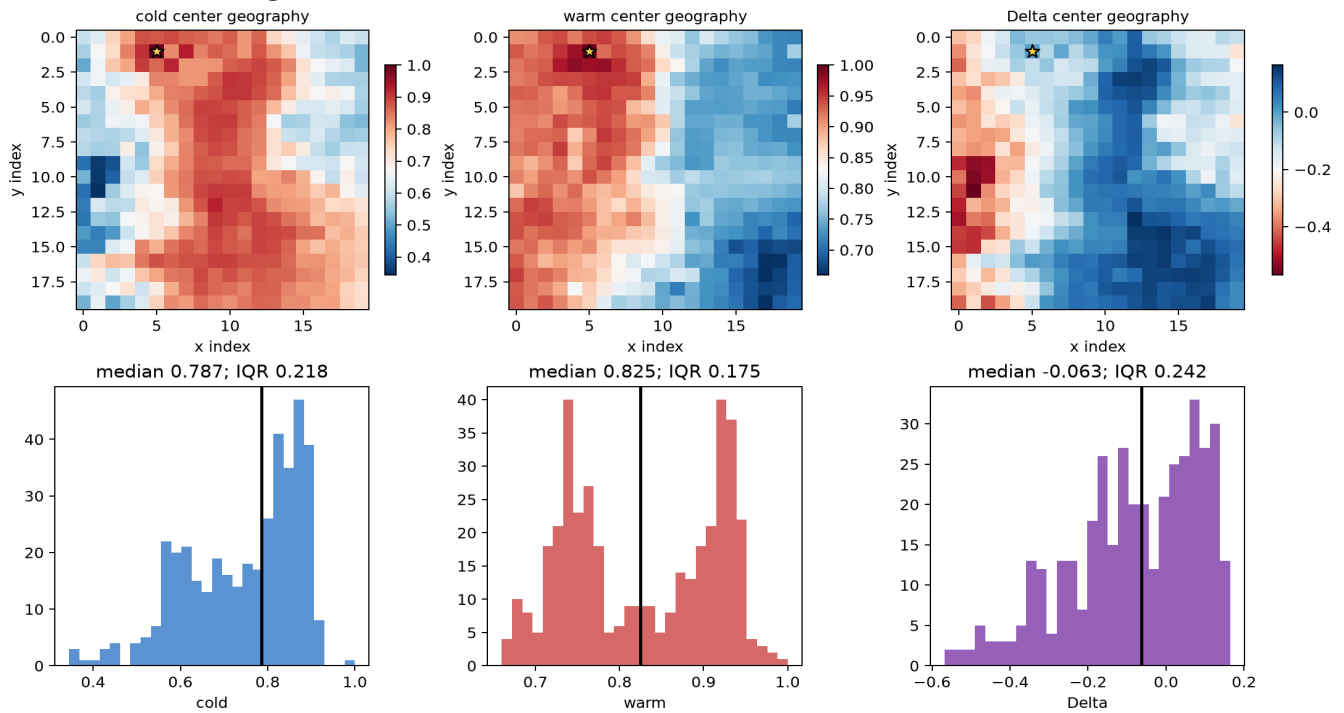
High-IQR Phase 1 focal: broad, continuous, directional



This broad stack is KDE-unimodal at all three tested bandwidths. Distance-bin eta-squared is about 0.10, direction eta-squared about 0.29, and distance residual IQR remains about 0.185. It is best described as broad continuous, directionally organized structure—not a validated multimodal transition.

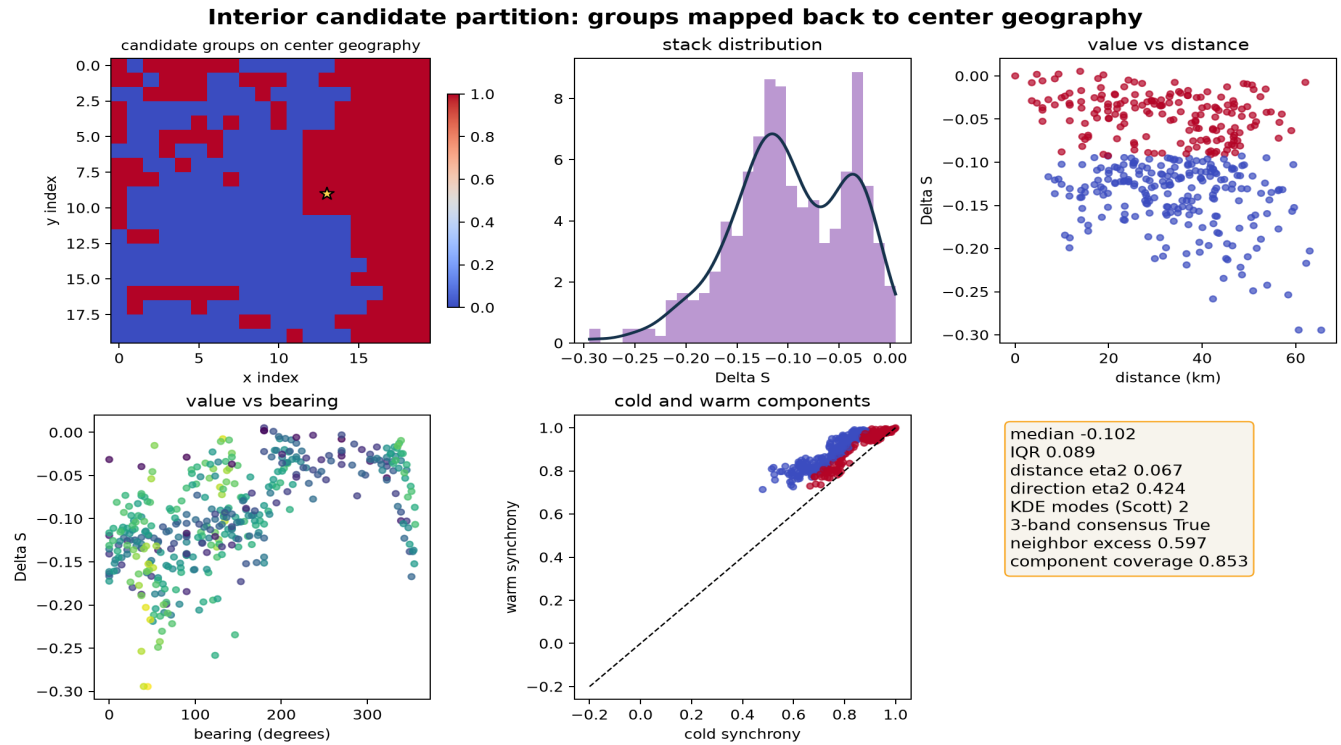
9. Cold/warm decomposition

High-IQR focal (1,5): cold and warm contributions remain distinct



At the high-IQR focal, cold IQR (about 0.218) exceeds warm IQR (about 0.175), but both components vary. Delta cannot be interpreted without preserving both inputs; the Delta median remains the median of pairwise cold-minus-warm values.

10. Candidate spatial partition



The interior focal (9,13) retains two KDE modes across 0.75x, 1.0x, and 1.25x Scott bandwidth. Its candidate groups occupy coherent center geography rather than being randomly intermixed. Across the block, 22 of 400 focals meet this bandwidth-consensus screen, many near margins. This is candidate structure only: one season, bounded support, forced two-group labels, and edge confounding preclude a climate-regime claim.

11. Stack typology

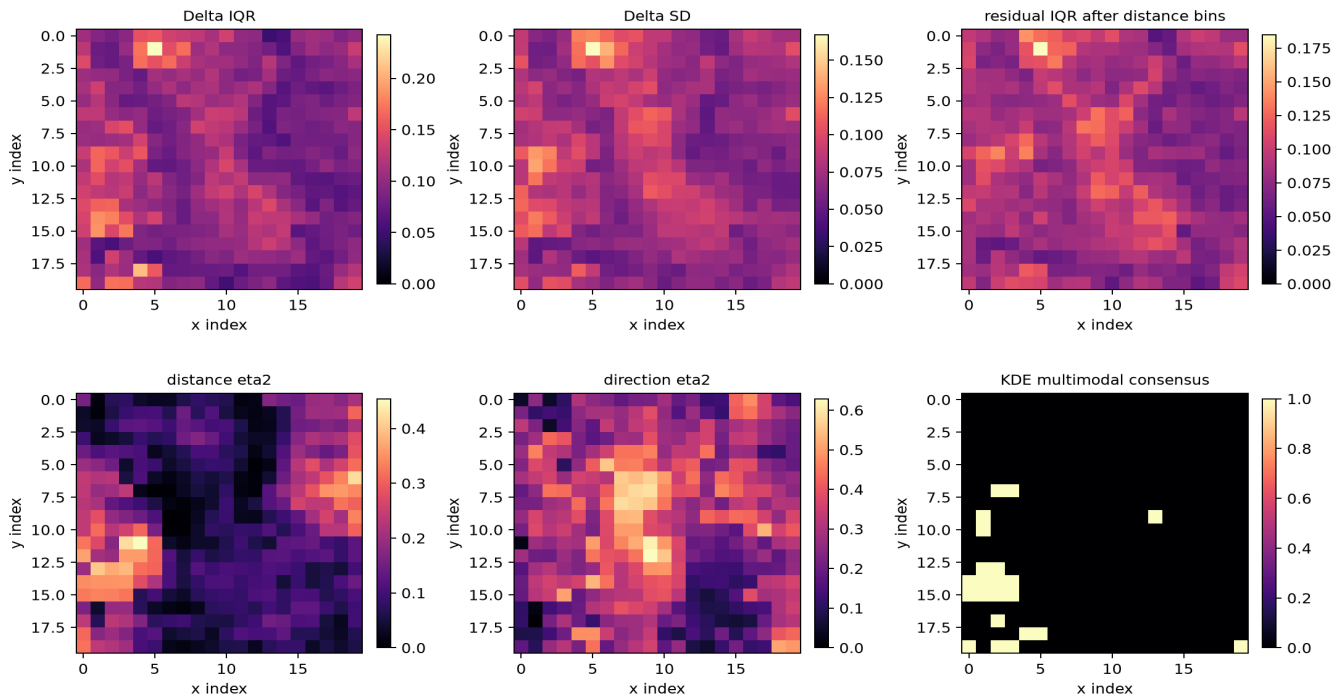
All six requested descriptive types have real examples in this block, but the names are selection roles rather than inferred climate classes.

Type	Focal	IQR	distance eta2	direction eta2	modes	consensus
Type 1 - low variance / spatially uniform	(17,5)	0.056	0.076	0.124	1	False
Type 2 - distance structured	(11,4)	0.104	0.454	0.287	1	False
Type 3 - directionally structured	(12,9)	0.129	0.058	0.629	1	False
Type 4 - broad continuous	(1,5)	0.242	0.104	0.290	1	False
Type 5 - candidate spatial partition	(9,13)	0.089	0.067	0.424	2	True
Type 6 - edge / incomplete support	(0,0)	0.109	0.158	0.048	1	False

Standardized ten-panel figures are written as separate reusable PNG artifacts and included in the appendix.

12. Heterogeneity and organization

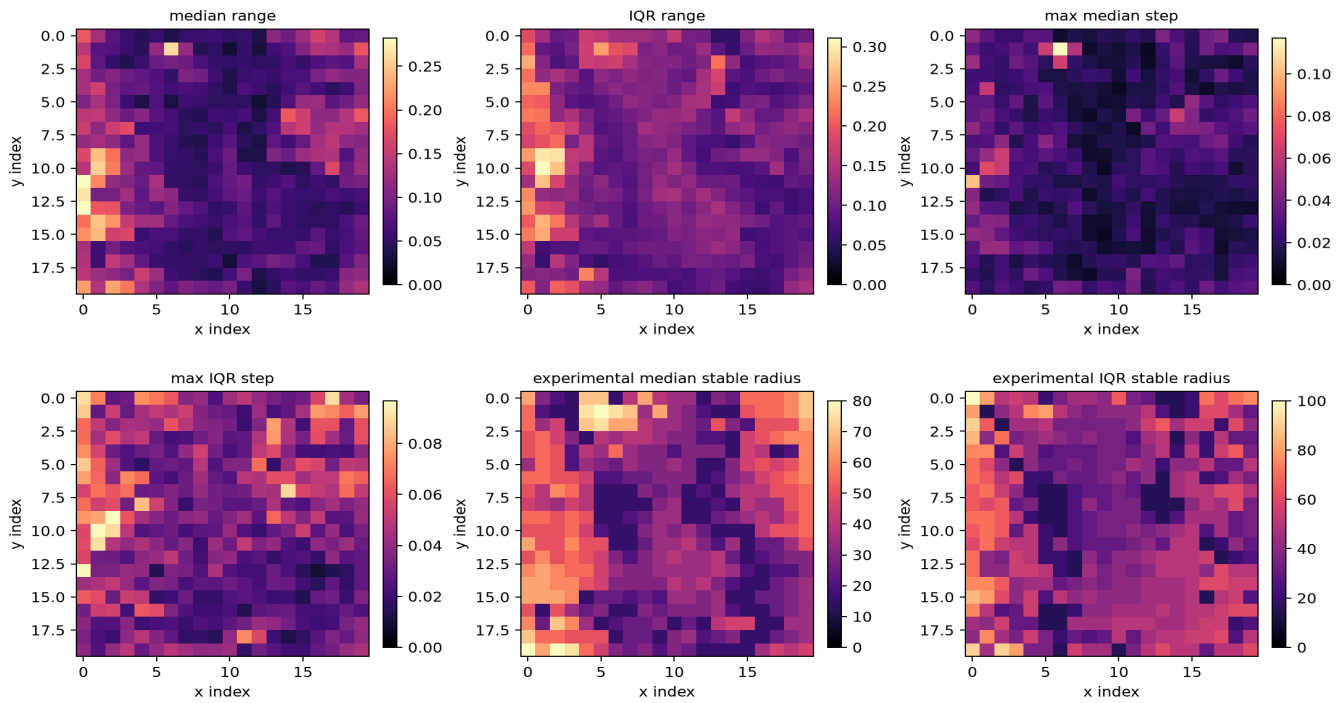
Heterogeneity, distance, direction, and modality are not one map



High IQR is neither multimodality nor spatial partitioning. Distance, direction, residual spread, and bandwidth-stable modes preserve different evidence.

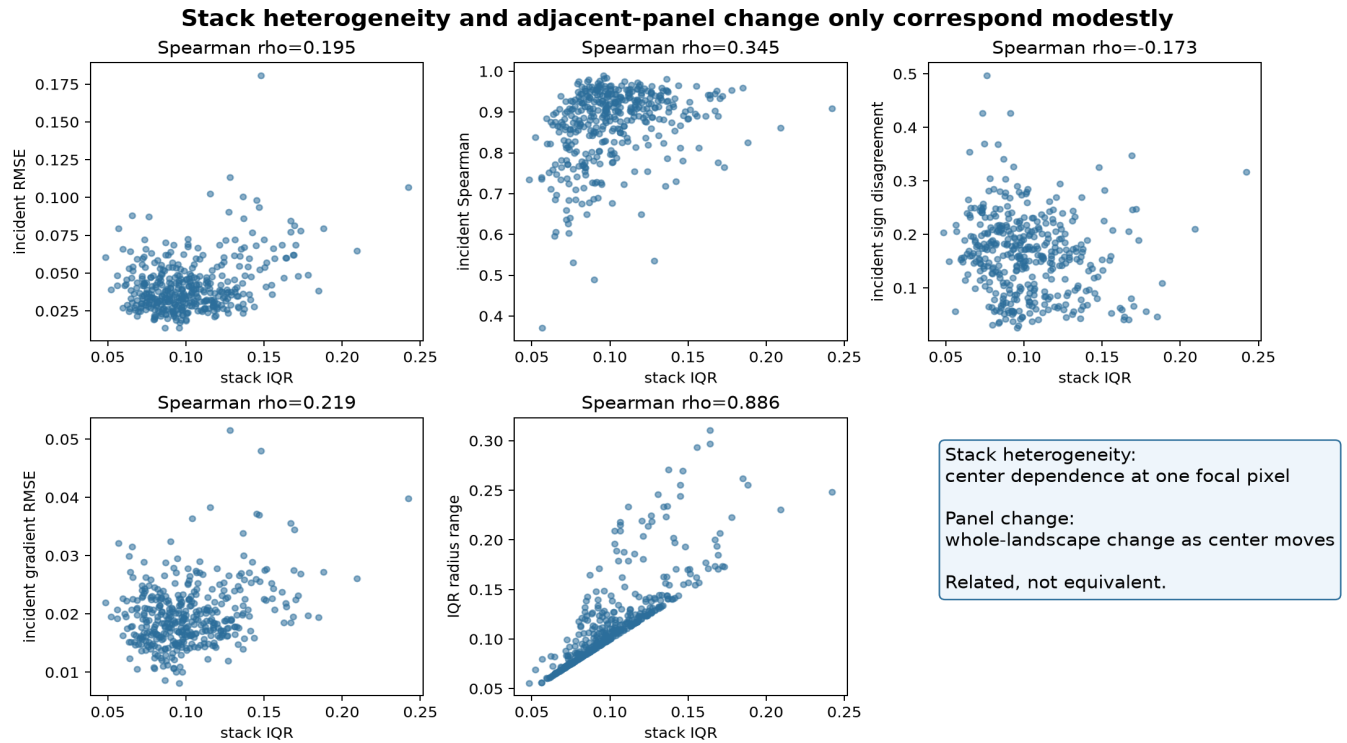
13. Radius sensitivity

Support-radius sensitivity is strongly spatially variable



Experimental stable median radii span 15-80 km (median 35 km); IQR radii span 15-100 km (median 40 km). Delta median radius ranges span about 0.027-0.282; IQR ranges span about 0.056-0.311. One fixed support radius is not equally representative everywhere in this block.

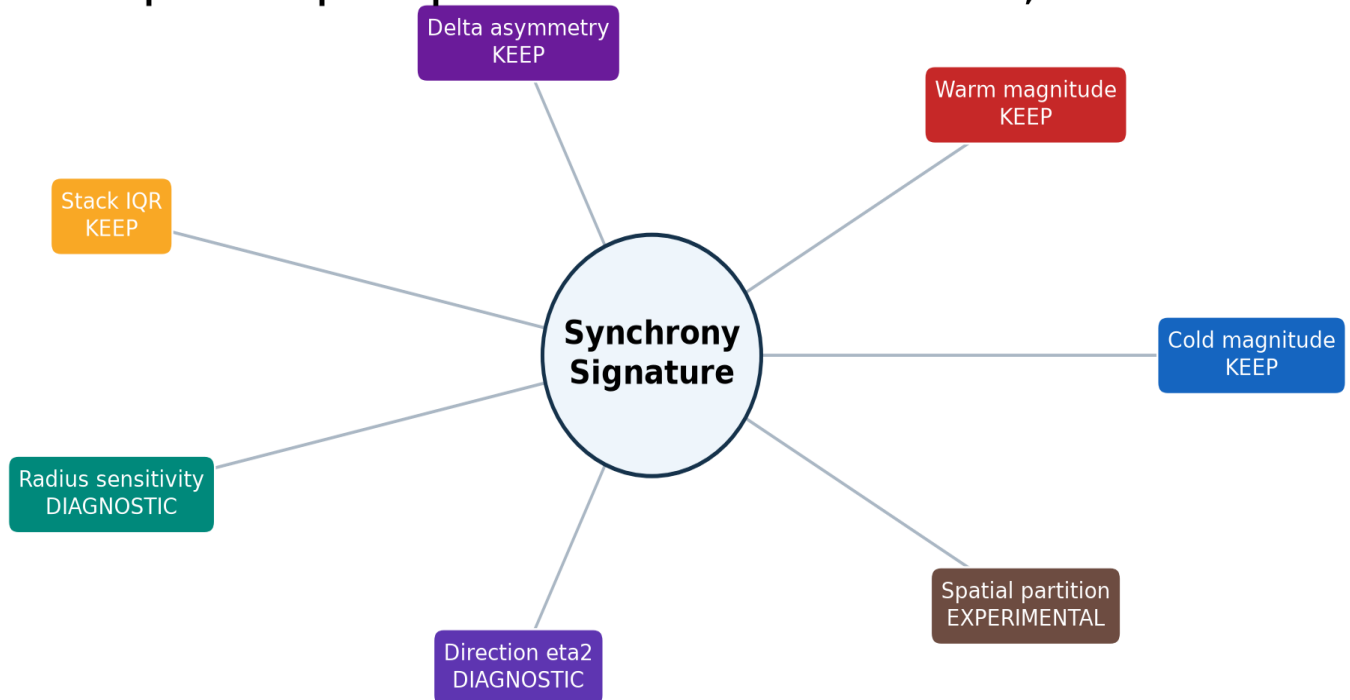
14. Heterogeneity versus panel change



Correspondence is modest and sometimes counterintuitive. Stack IQR versus incident RMSE is positive but not strong; normalized RMSE and sign change can be weakly negative. The two objects answer different questions and must not be substituted for each other.

15. Proposed SynchronySignature

Proposed compact representation: evidence-ranked fields, not one scalar



The candidate schema is machine-readable. Robust magnitude and IQR fields are ready; radius and direction are diagnostic; spatial partitioning remains experimental. No universal scalar preserves all axes.

16. Phase 1.5 decision table

Diagnostic	Scientific question	Mathematical definition	Robustness in Phase 1.5	Redundant with	Computational cost	Nationally scalable?	Interpretability	Recommendation
cold/warm/Delta median	magnitude and tail asymmetry	median over pair values by focal pixel	stable, directly interpretable	none	streamable quantile	yes with bounded support	high	KEEP
Delta IQR	center dependence / heterogeneity	q75-q25 over focal stack	large spatial contrast; robust	SD partially	streamable/sketchable quantile	yes	high	KEEP
Delta SD	stack dispersion	standard deviation over centers	similar role to IQR; tail sensitive	IQR	online moments	yes	high	KEEP AS DIAGNOSTIC
panel Spearman	rank-pattern change	rank correlation of adjacent landscapes	near ties can destabilize ranks	none	panel materialization/ranking	sample/tile boundaries	conditional	KEEP AS DIAGNOSTIC
panel normalized RMSE	relative magnitude change	RMSE / pooled q95-q05	complements absolute RMSE	RMSE partly	linear in panel pixels	yes by neighboring centers	high with denominator	KEEP

Decision table (continued)

Diagnostic	Scientific question	Mathematical definition	Robustness in Phase 1.5	Redundant with	Computational cost	Nationally scalable?	Interpretability	Recommendation
panel sign disagreement	tail-asymmetry class change	fraction of deadbanded signs unequal	deadband avoids zero noise	none	linear	yes	high	KEEP
panel gradient RMSE	spatial reorganization	RMSE of x/y gradient-vector differences	useful but grid-resolution dependent	RMSE correlated	linear	yes	moderate	KEEP AS DIAGNOSTIC
Wasserstein distance	distribution shift only	1D transport distance between panel values	ignores spatial arrangement; RMSE correlation high	RMSE/MAE	sorting	yes	moderate	DROP
radius ranges/steps	support sensitivity	range and maximum step across nested radii	strongly spatially variable	none	nested accumulators	yes at fixed rings	high	KEEP AS DIAGNOSTIC
experimental stable radius	descriptive convergence scale	first supported radius with remaining range ≤ 0.02	threshold dependent	radius ranges	small after ring summaries	yes	conditional	EXPERIMENTAL

Decision table (continued)

Diagnostic	Scientific question	Mathematical definition	Robustness in Phase 1.5	Redundant with	Computational cost	Nationally scalable?	Interpretability	Recommendation
distance eta2 / bins	distance-associated variation	between-bin variance / total variance	descriptive, bin dependent	distance Spearman partly	ring summaries	yes	high	KEEP AS DIAGNOSTIC
direction eta2	directional organization	between-bearing-bin variance / total variance	clear signal; block sensitive	none	8-bin summaries	yes	high	KEEP AS DIAGNOSTIC
KDE modes + coherent groups	candidate spatial partition	3-bandwidth modes plus mapped 2-group coherence	22 candidates, edge confounding remains	none	requires retained/sample stacks	not yet	experimental	EXPERIMENTAL

17. Required answers and final gate

1. Why Q about 0.088?

Narrow dynamic range and dense near-ties make ranks unstable; there is also real cold-driven local/sign change. RMSE 0.080 is modest absolutely but 0.429 of the pooled robust range.

2. Is Spearman useful?

Yes, as a rank-structure axis with range/tie context; not as a universal boundary scalar.

3. What accompanies it?

Normalized RMSE, deadbanded sign disagreement, and gradient-vector RMSE; retain cold/warm/Delta decomposition.

4. Do stacks stabilize?

Many do descriptively, but curves can be nonmonotonic and stabilization is not uniform.

5. At what scales here?

Experimental median stable radius: 15-80 km, median 35 km. IQR: 15-100 km, median 40 km.

6. Spatially variable?

Yes, strongly.

7. Fixed radius defensible?

Not as a universal scientific radius from this evidence. A common benchmark cap is defensible only with nested-ring sensitivity.

8. Distance explanation?

Distance-bin eta-squared has median about 0.10 and maximum about 0.45; distance alone usually leaves substantial residual spread.

9. Direction evident?

Yes. Direction eta-squared has median about 0.285 and reaches about 0.629, while remaining a descriptive block-sensitive statistic.

10. High-IQR modes?

The Phase 1 high-IQR example is broad continuous and bandwidth-stable unimodal. A minority of other stacks are candidate multimodal.

11. Modes geographically organized?

For the 22 screened candidates, mapped two-group labels can be strongly coherent; many candidates are near block margins, so none establishes a regime.

12. High-IQR transition?

No: it is not a multimodal transition under these tests; it is broad and directionally organized.

13. Cold/warm contributions?

Both vary, with larger cold than warm spread at the high-IQR focal. The strongest panel-rank break is mainly cold-driven while warm ranks remain stable.

14. Same phenomenon?

No. Associations are modest; focal center-dependence and whole-landscape transition are distinct.

15. Smallest nonredundant set?

Cold median, warm median, pairwise-Delta median, Delta IQR, nested radius sensitivity, direction eta-squared, and the four-axis panel-change signature. Partition fields remain experimental.

16. National maps?

Cold, warm, Delta medians; Delta IQR; bounded-radius sensitivity; and selected panel-change axes after the benchmark.

17. Diagnostics only?

SD, distance/direction profiles, experimental stable radius, KDE modes/group coherence, Pearson, and gradient details beyond the retained transition axis.

18. Preserve before discarding pairs?

Cold/warm/Delta values and valid counts by bounded edge or exact mergeable summaries; distance/bearing; nested-ring accumulators/quantile sketches; adjacent-center panels or sufficient tile halos; sampled full stacks for audit.

19. Phase 2 radius strategy?

Benchmark nested 25, 50, 75, and 100 km rings with halos; compare convergence and edge bias rather than selecting one radius beforehand.

20. Ready for 100 by 100?

GO WITH CHANGES: yes for a bounded observed benchmark with nested support and retained diagnostics; no for a national run or regime product.

18. Phase 2 bounded benchmark contract

Accumulate: separate cold, warm, and Delta pair values, joint-tail counts, distance and bearing, and canonical pair identity. Retain: medians, Delta IQR, nested-ring summaries, panel normalized RMSE/Spearman/sign/gradient signature, audit samples, fingerprints, and exact support metadata. Test: 25/50/75/100 km with tile halos and explicit edge checks. Discard only after equivalence: redundant MAE/Wasserstein fields and complete dense pair arrays outside audit samples. Experimental: stable-radius threshold, KDE modes, forced groups, and any regime label.

Graph interpretation

Pair computation and storage benefit from an undirected sparse graph. A focal stack is a node's incident edges; a center landscape is those edges rasterized by the other endpoint. Panel comparison compares two such edge-pattern rasters. Distance rings, bearing bins, neighborhood coherence, and final products remain spatial-raster operations because that form preserves geography and support.

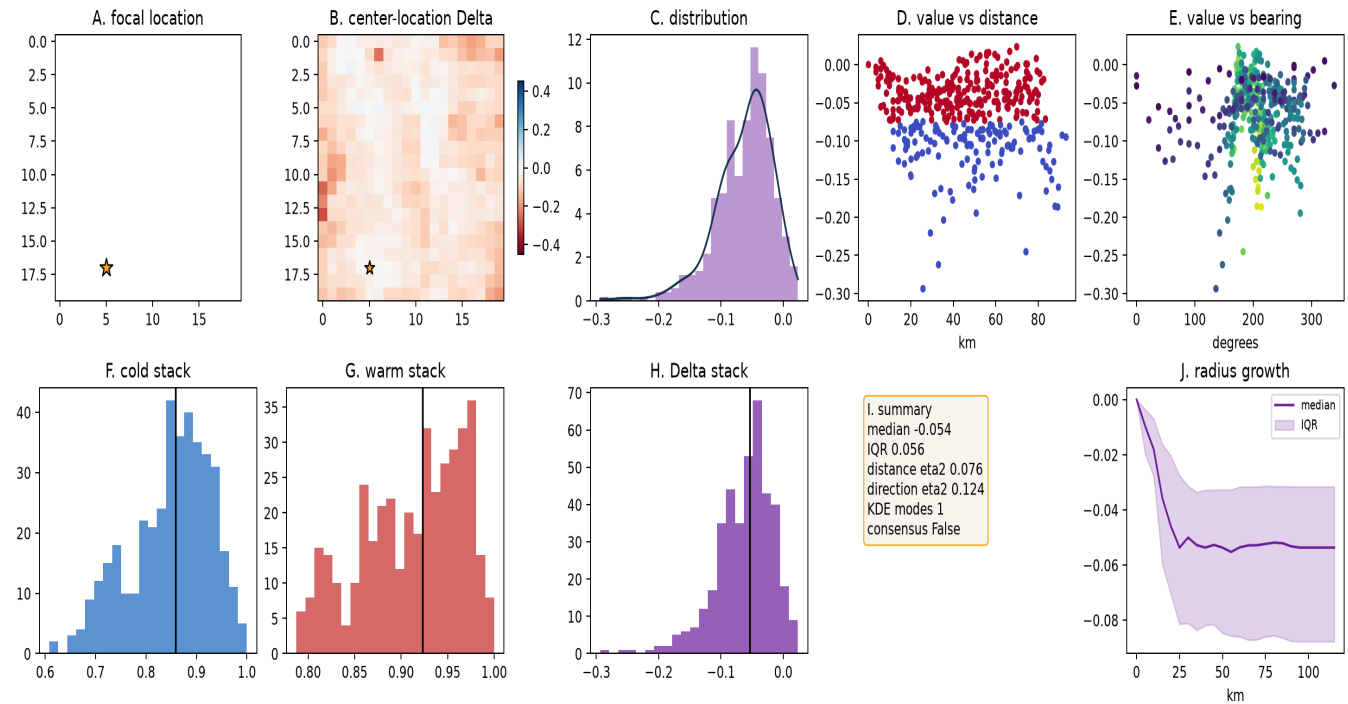
Limitations

One observed Colorado winter block cannot establish seasonal or national invariance. The 20 by 20 domain truncates support and can create edge-associated structure. Eta-squared depends on bins, gradient metrics depend on resolution, and the stability threshold is exploratory. KDE consensus and spatially coherent forced groups flag candidates; they do not identify causal or climatological regimes.

Appendix: standardized typology figures

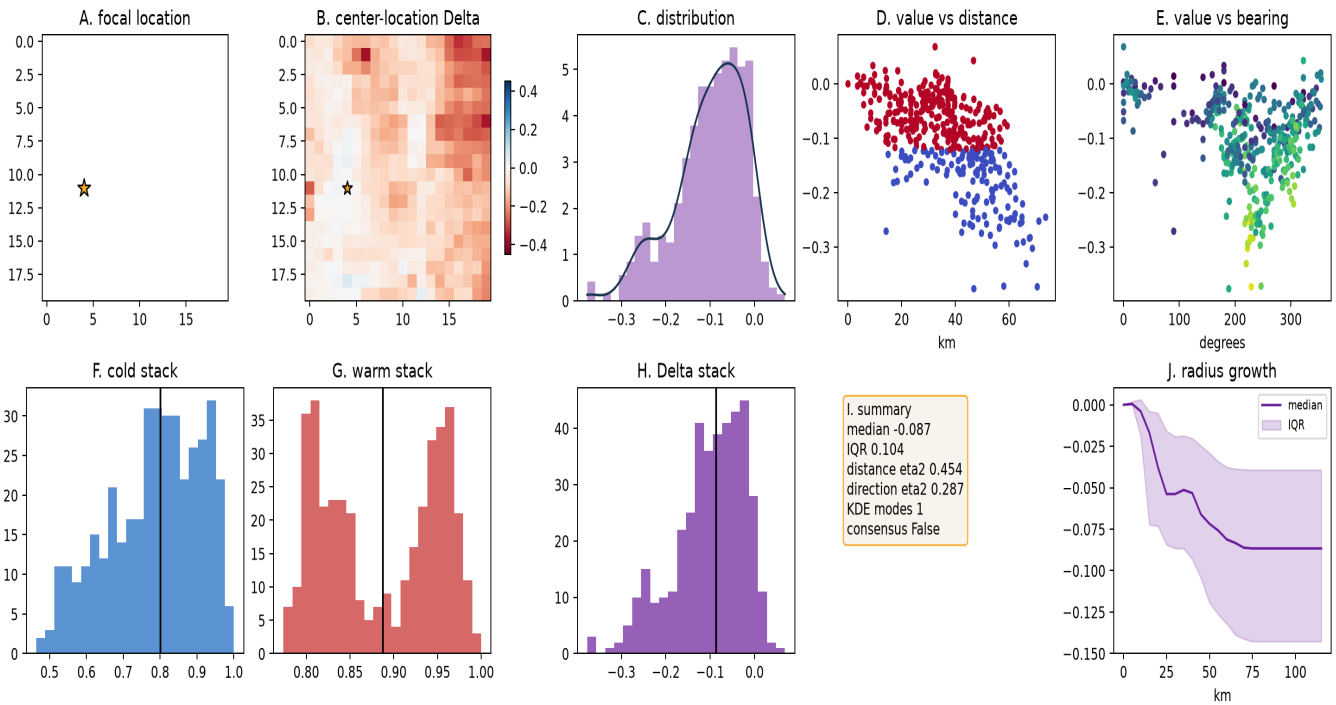
Type 1 - low variance / spatially uniform

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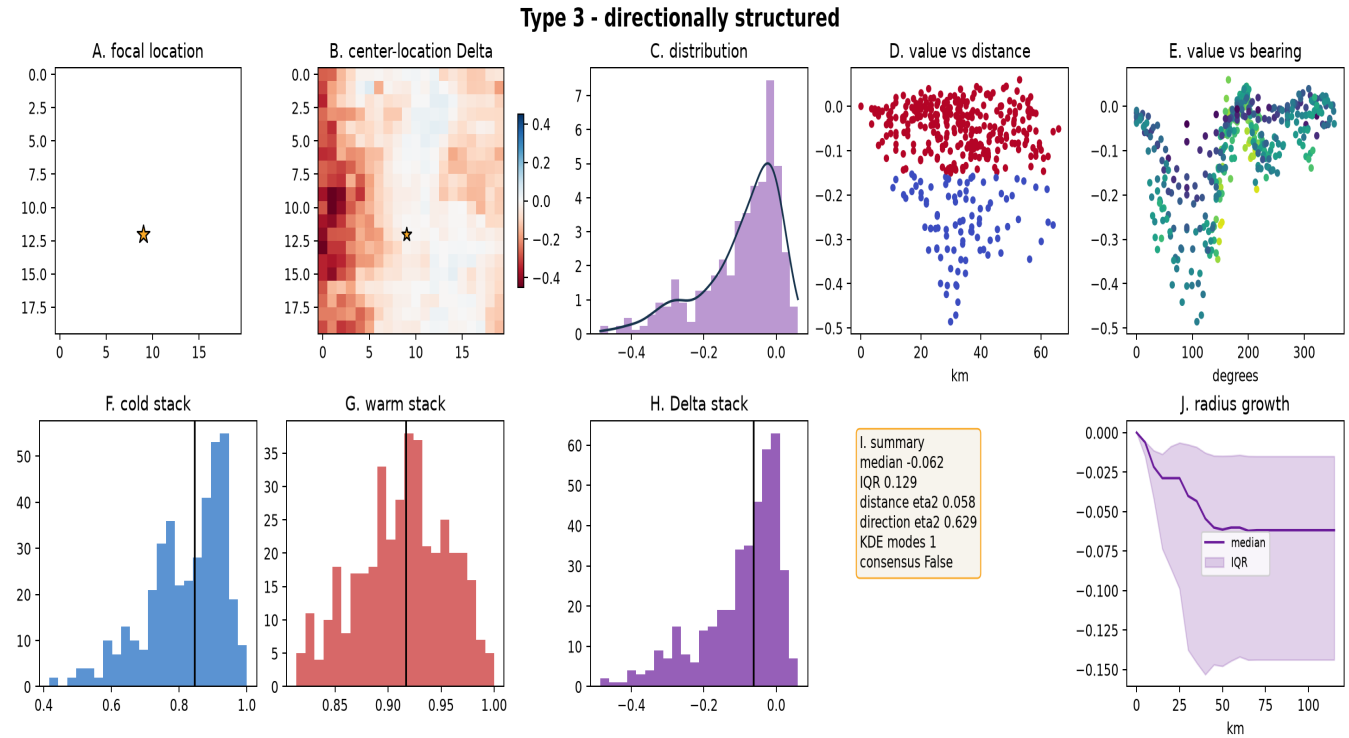


Type 2 - distance structured

Type 2 - distance structured

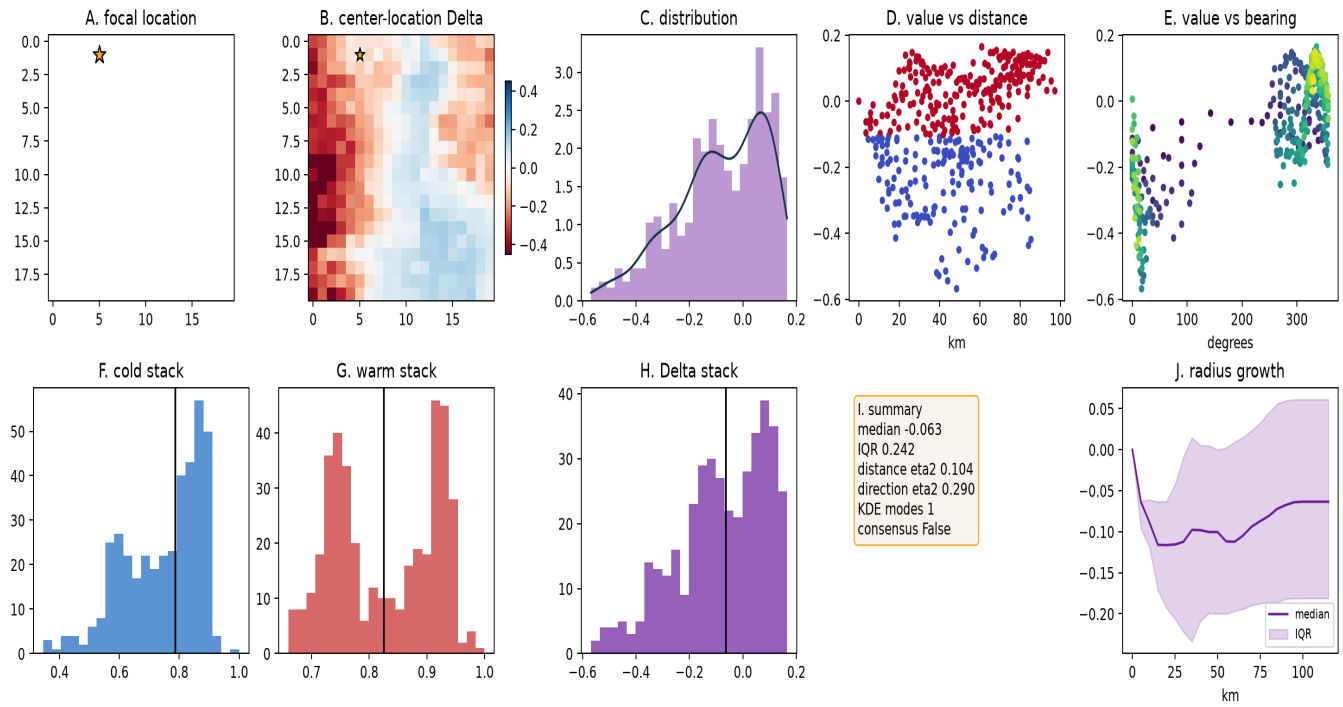


Type 3 - directionally structured



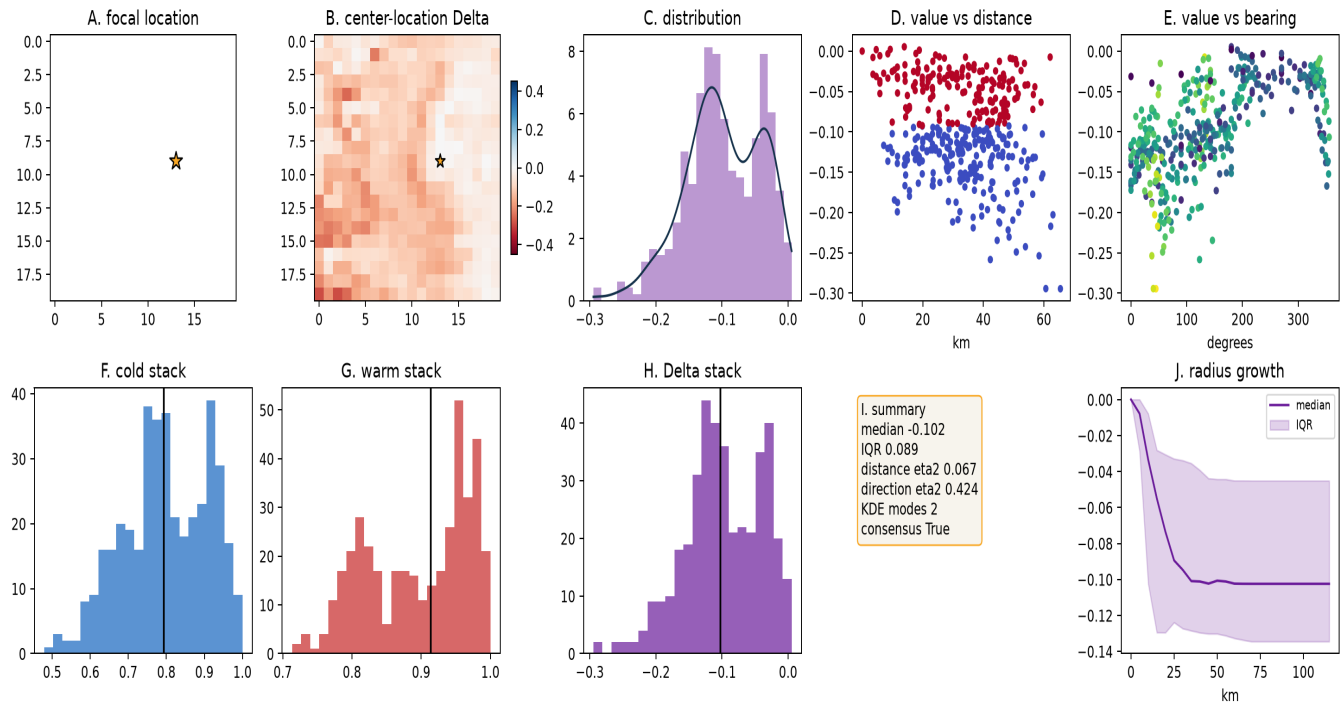
Type 4 - broad continuous

Type 4 - broad continuous



Type 5 - candidate spatial partition

Type 5 - candidate spatial partition



Type 6 - edge / incomplete support

Type 6 - edge / incomplete support

